

Section 15.2

1. Evaluate $\iint_D (x + y) dA$, where D is the triangular region with vertices $(0, 0)$, $(2, 0)$, and $(0, 4)$.

Hint: Sketch the region. The hypotenuse has equation $y = 4 - 2x$. Set this up as a Type 1 integral.

2. Evaluate $\iint_D x dA$, where D is the region bounded by $x = y^2$ and $x = 4$.

Hint: Sketch the region. For a fixed y , where does x start and stop? What are the y -bounds?

3. Evaluate $\iint_D 2x \, dA$, where D is the region bounded by $x = y^2 - 1$ and $x = 3 - y^2$.

Hint: Sketch the region. Can you describe it as Type 1? As Type 2? Only one of these works as a single integral.

4. The integral $\int_0^4 \int_{\sqrt{y}}^2 \cos(x^3) \, dx \, dy$ cannot be evaluated directly because $\cos(x^3)$ has no elementary antiderivative in x .

Reverse the order of integration and evaluate.

Hint: Sketch the region of integration first. The inner limit $x = \sqrt{y}$ is the same as $y = x^2$.